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import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Load dataset
data = pd.read_csv("dataset/loan_prediction.csv")

print("Original Missing Values:")
print(data.isnull().sum())

# Fill categorical columns
categorical_cols = ["Gender", "Married", "Dependents", "Self_Employed"]

for col in categorical_cols:
    data[col] = data[col].fillna(data[col].mode()[0])

# Fill numerical columns
numerical_cols = ["LoanAmount", "Loan_Amount_Term", "Credit_History"]

for col in numerical_cols:
    data[col] = data[col].fillna(data[col].mean())

print("\nMissing Values After Filling:")
print(data.isnull().sum())

# Encode categorical columns
encoder = LabelEncoder()

columns = [
    "Gender",
    "Married",
    "Dependents",
    "Education",
    "Self_Employed",
    "Property_Area",
    "Loan_Status"
]

for col in columns:
    data[col] = encoder.fit_transform(data[col])

# Drop Loan_ID
data.drop("Loan_ID", axis=1, inplace=True)

# Save cleaned dataset
data.to_csv("dataset/cleaned_loan_data.csv", index=False)

print("\nFirst 5 Rows:")
print(data.head())

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print("\nCleaned dataset saved successfully!")
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